IT University of Copenhagen

Building the Democracy Stack

What Software Architecture Decisions Matter for Democratic Digital Infrastructure?

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Abstract

Civic technology platforms have been building digital democratic infrastructure for over a decade — software for participatory budgeting, policy proposals, and citizen consultations. European policy is now converging on the same idea, under names like the Digital Services Act, the European Digital Identity framework, IDEA's "democracy stack", and CEPS's case for European digital public infrastructure. But policy talks about what this infrastructure should do, not how it should be built. The platforms themselves have rarely been analysed through software architecture and software ecosystem (SECO) methods. This paper proposes a small analytical framework for reasoning about software architecture decisions in democratic digital infrastructure. The framework has three components: four quality attributes that connect architecture to democratic capacities (modifiability, interoperability, configurability, and transparency and auditability); the three-structures lens of Christensen et al. 2014, where ecosystem health depends on the fit between organisational, business, and software structures; and a set of architectural trade-offs derived from the four attributes. Alongside these, the paper adapts fork-mining analysis into a population-shape niche-creation diagnostic, *fork-divergence shape*, for civic tech ecosystems.

The framework is applied to a comparative case study of Decidim and Consul, two civic tech platforms with shared origins in Spain's 15M movement and opposite architectural paths. The application uses architectural recovery, evolutionary analysis, and a fork-divergence study across all public forks of both platforms (1,030 for Consul, 452 for Decidim). Decidim's modular software fits its coordinating Association and consultancy-anchored funding base; Consul's monolithic software requires a coordinating capacity that its small Foundation and shoestring funding cannot provide. The fork-divergence shapes invert at the originator: Madrid (Consul) sits in an abandoned 12,725-commit silo, Barcelona (Decidim) tracks upstream with zero local commits. On each platform the diagnostic reads a different niche — deployment fragmentation for fork-based Consul, contributor flow for plugin-based Decidim — bounded by the platform's extension model. The two-case scope means that quality attributes such as security, accessibility, and sovereignty, which Decidim and Consul do not strongly differentiate, are left for future work.

Contents

1 Introduction

European policy increasingly names digital infrastructure for democracy — the Digital Services Act, the European Digital Identity framework, IDEA's "democracy stack" [International IDEA, 2026], and CEPS's case for European Digital Public Infrastructure [CEPS, 2025]. These policy proposals name what they want, not what the software architecture should look like. Civic technology platforms have been building this kind of software for over a decade — Decidim across hundreds of institutions in 32 countries [Barandiaran et al., 2024], Consul across over 250 cities and organisations [NTARI, 2025] — yet how their architectures shaped their ecosystems remains under-analysed.

This paper asks: *what software architecture decisions matter for democratic digital infrastructure, and how can they be analysed through software architecture and software ecosystem methods?*

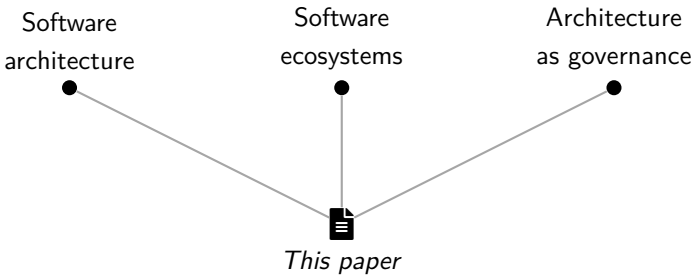
The methods exist across software architecture, SECO, STS/platform studies, and fork-mining research (Section 2), but have rarely been combined to analyse civic tech: Manikas [2016]'s 2016 review of 231 SECO studies found no work on civic or commons-based platforms.

We propose a small analytical framework combining four quality attributes (modifiability, interoperability, configurability, transparency and auditability), Christensen et al. [2014]'s three-structures lens, and a set of trade-offs. We apply it to Decidim and Consul — two platforms with shared origins in Spain's 15M movement (2015–2016) but opposite architectural paths (Decidim modular, 27 Ruby engines + coordinating Association; Consul monolithic, deployers fork + small Foundation coordinates from the outside) — using architectural recovery, commit history analysis, and a fork-divergence study across 1,030 Consul and 452 Decidim public forks. From the application we surface *fork-divergence shape*, a population-shape niche-creation diagnostic whose reading depends on the extension model. Decidim and Consul are first worked cases, not a full validation across the field.

Section 2 reviews the literature; Section 3 describes the method; Section 4 presents the framework; Section 5 applies it to Decidim and Consul; Section 6 discusses what the framework shows and does not; Section 7 concludes.

2 Background

Software architecture research gives us quality attributes and trade-offs as a way of reasoning about design decisions. Software ecosystem research gives us a way of analysing the organisational, business, and technical structures around a platform. Architecture-as-governance literature shows that software architectures participate in governance by making some forms of action, coordination, and control easier than others. We need all three to reason about democratic digital infrastructure.



Three literatures combined into one analytical framework.

2.1 Software Architecture Quality Attributes

Bass et al. [2021, p. 4] call software architecture "the set of structures needed to reason about a system." Architecture decisions affect a system's properties over its lifetime through quality attributes: measurable properties like modifiability, performance, security, and so on. Quality attributes are useful here because they let us evaluate architecture without reducing it to implementation details. They allow us to ask not only what a platform contains, but what kinds of change, inspection, integration, and adaptation its structure makes easier or harder.

In commercial software, quality attributes are usually optimised for business value. For democratic digital infrastructure the goal is different: which architectural properties support democratic participation, public legitimacy, and institutional adaptability?

We focus on four quality attributes.

Modifiability. How easily the system can be changed. Civic tech platforms have to adapt to communities with different rules, languages, and participatory traditions; modifiable platforms can do this without forcing each deployer to maintain a separate fork.

Interoperability. How well the system exchanges data and services with others. This matters because no single platform will serve all democratic functions, so platforms need to compose.

Configurability. The platform's support for configuring varied participation processes (proposals, voting, deliberation, accountability) without changing the source code. PolicyKit [Zhang et al., 2020] prototyped programmable governance for online communities, a close analogue of this property at the governance-rule layer.

Transparency and auditability. How much of the platform's operation can be inspected by relevant observers (transparency), and how reliably those observations can be used to verify and contest outcomes (auditability). Democratic legitimacy depends partly on both, although technical auditability does not by itself guarantee that citizens can meaningfully use these mechanisms in practice.

Other quality attributes (security, accessibility, sovereignty, performance) matter for democratic infrastructure too. We do not treat them as primary axes here and return to this scoping in Section 6.

2.2 Software Ecosystems

SECO research broadens the unit of analysis from the software to the wider system around it. Manikas and Hansen [2013] define a software ecosystem (SECO) as "the interaction of a set of actors on top of a common technological platform, which results in a number of software solutions or services." The concept extends beyond a single product to the organisations, developers, and users that develop around a shared platform.

Christensen et al. [2014] propose that ecosystems can be analysed through three interacting structures: an *organisational* structure (who participates, in what roles, how they coordinate), a *business* structure (who pays, who captures value, what incentives flow), and a *software* structure (the platform architecture, extension mechanisms, APIs). Their argument is that ecosystems thrive when these three structures fit together and struggle when they do not. We use this lens in Section 5 to compare Decidim and Consul.

We also draw on Iansiti and Levien [2004]'s ecosystem actor classification (keystones, dominators, niche creators) and Jansen [2014]'s ecosystem health dimensions (productivity, robustness, niche creation). In this paper, these concepts are treated as analytical lenses rather than direct measurements of democratic value.

Manikas [2016]'s systematic review of 231 SECO studies found none on commons-based, civic, or democratic platforms. Knutas et al. [2020] proposed applying OSSECO methods to civic tech but did not carry out the empirical work. Palacin et al. [2024] study Decidim's modularity from a human-computer interaction perspective, but do not analyse the architecture itself as part of a software ecosystem.

2.3 Architecture as Governance

A long tradition in science and technology studies (STS) and platform studies has established that software architectures are not politically neutral: they distribute possibilities, making some forms of participation, oversight, and control easier than others [Winner, 1980, Lessig, 2006, DeNardis, 2014, Schneider, 2024]. This thesis is analytical rather than deterministic — architecture does not by itself cause political outcomes. Schneider et al. [2021]'s *Modular Politics* moves toward design, proposing four principles (modularity, expressiveness, portability, interoperability) for a governance layer in online communities. Two of these principles overlap with QAs developed in this paper (modularity with modifiability, interoperability with interoperability), and we revisit

this resonance in Section 4. These ideas are directly relevant to civic technology platforms, but have rarely been combined with software architecture and SECO methods.

Three literatures, each well-developed on its own but rarely combined for civic tech. Section 3 describes the method that combines them.

3 Method

This paper is a comparative case study of two civic technology platforms, Decidim and Consul, following the case-study guidelines for software engineering of Runeson and Höst [2009]. The empirical work draws on two methodological traditions: architectural recovery [Ducasse and Pollet, 2009, Garcia et al., 2013] for reconstructing each platform's software structure from primary sources, and Mining Software Repositories [Hassan, 2008] for the commit-history, contributor-concentration, logical-coupling, and fork-divergence analyses. The framework presented in Section 4 is an analytical lens synthesised from existing software architecture and SECO literature; the application in Section 5 demonstrates it on the two cases but does not validate it across the wider civic tech field. We therefore present the framework and apply it to one comparison, leaving evaluation across more cases as future work.

Case selection. We selected Decidim and Consul as the comparative case for two reasons. First, both have publicly developed codebases, active deployments, and enough documentation for architectural recovery. Second, the two platforms share an origin in Spain's 15M movement in 2015–2016 but took opposite architectural paths: Decidim went modular (27 Rails engines, gem-based extension); Consul stayed monolithic (single Rails app, fork-based extension). The shared origin reduces some of the variance that usually confounds case comparisons in software studies, while the architectural divergence provides the contrast we want to study. This is theoretical case selection: the cases are chosen for their analytical contrast, not statistical representativeness. Other civic tech platforms (Polis, Loomio, CitizenLab, vTaiwan, Adhocracy) are not analysed; a single two-platform comparison is enough to demonstrate the framework at this stage, but not enough to generalise across civic tech as a whole.

Architectural recovery. For each platform we recover the software structure from primary sources: codebase organisation, gemspec files, extension mechanisms, and documentation. For Decidim, a script parses all 27 first-party gemspecs at a pinned snapshot, extracts every `add_dependency` call, classifies each as internal or external, and identifies `Decidim.register_component` and `Decidim.register_participatory_space` calls. The engine dependency graph in Section 5 is recovered from this data, not asserted from documentation. Consul's recovery is structural: a Rails monolith with a file-override extension mechanism. The recovery captures dependency structure and extension mechanisms, not runtime behaviour or deployment-specific customisations.

Evolutionary analysis. For both platforms we extract commit metadata with `git log --no-merges` on partial clones. Bot authors (Dependabot, Crowdin, Weblate, GitHub Actions, named

release bots) are filtered by email and name patterns. We compute total human commits, unique authors, top-three and top-ten contributor share, Gini coefficient (a measure of contributor concentration: 0 means commits are evenly distributed across authors, 1 means one author makes them all), and the per-year commit timeline; for Decidim we also extract per-engine commit counts. We use these as indicators of ecosystem activity and concentration [Hassan, 2008, Manikas and Hansen, 2013, Jansen, 2014]. They are not treated as direct measures of ecosystem health or democratic value: commit counts show activity, not necessarily meaningful maintenance; author counts show participation, not necessarily influence; and concentration metrics show distribution, not governance quality.

Logical coupling. Following Gall et al. [1998], we measure which Decidim engines change together. For each non-merge non-bot commit we identify the engines touched and compute Jaccard similarity (commits touching both / commits touching either) for each engine pair. Mega-commits touching more than eight engines are excluded as refactor or release commits that would otherwise dominate the signal. Logical coupling is computed only for Decidim, since Consul's monolithic structure has no engine boundary to test against. This tests whether the manifest pattern is reflected in change isolation at the repository level, not only in the gemspec dependency structure.

Fork divergence. For both platforms we use the GitHub REST API to enumerate all public forks of the upstream repository (`consuldemocracy/consuldemocracy`, formerly `consul/consul`; and `decidim/decidim`) and call the compare endpoint for each, returning commits ahead and behind upstream's default branch (`master` for Consul, `develop` for Decidim), drawing on the fork-mining literature [Zhou et al., 2020, Wang et al., 2023, Businge et al., 2022]. Each fork is classified as *modified* or *unmodified* (commits ahead) and *active* or *dormant* (pushed in the last three years; this is a methodological choice rather than a fixed convention). The analysis covers public forks only and may miss private production forks. The two platforms have different extension models: Consul deployers customise by forking the application, while Decidim deployers install the gem in their own Rails application and extend through plugins. As a result, the same diagnostic reads different ecosystem niches in each case — deployment fragmentation for Consul, contributor flow for Decidim — a methodological point we return to in the cross-case synthesis and discussion.

Three-structures application. For each platform we describe the three structures from Christensen et al. [2014] — organisational, business, software — using primary documentation (Decidim Association governance documents, MetaDecidim assembly records, Consul Democracy Foundation website, OSOR case studies) and the empirical material from architectural recovery and evolutionary analysis. We then assess alignment or misalignment between the three structures. This assessment is interpretive: it asks whether the software structure appears to afford the organisational and business reality around it, and whether the organisational and business structures appear able to sustain the architectural costs. Specifically, alignment evidence includes whether the organisational structure has roles to use the software's extension mechanisms, whether the business structure can sustain the architectural maintenance costs, and whether contribution

flows match the paths the software allows. It does not establish causal proof.

Quality-attribute assessment. The four quality attributes are assessed qualitatively as low, moderate, or high. Table 1 gives the criteria per QA per level. The ratings are based on observable architectural mechanisms and supporting evidence from the repository and documentation; they are structured judgements, not measurements.

Table 1: Quality-attribute rating rubric: criteria for low, moderate, and high ratings.

Quality attribute	Low	Moderate	High
Modifiability	Extension requires core code changes or a deployment fork	Plugin hooks exist; some adaptation still requires forking	Modules extend the platform; no core changes or forks needed
Interoperability	No public API or shared data formats	Partial API exposure or limited standard-format support	Stable public API; standard formats supported (e.g., ActivityPub, OpenAPI)
Configurability	Participation processes fixed in code	Some processes configurable; limited coverage	Varied processes composable via settings (proposals, voting, deliberation, accountability)
Transparency and auditability	Operations not logged beyond standard application logs	Some inspection mechanisms; not first-class architectural concerns	First-class logging, versioning, permission-gated visibility, public inspection

Data sources. Primary sources are the public GitHub repositories at pinned snapshots: `decidim/decidim` at `be39c244` (2026-05-12) and `consuldemocracy/consuldemocracy` at `3c3b5c4a` (2026-05-11). Secondary sources include Barandiaran et al. [2024] on Decidim's sociotechnical design, the OSOR case study on Consul's European deployments [Linåker and Gates, 2025], the NTARI fork report [NTARI, 2025], and the Codegram architecture blog on early Decidim design [Codegram, 2024]. Reproducibility scripts (`scripts/` directory: gemspec parsing, commit metadata extraction, fork-divergence analysis) and findings logs accompany this paper in a public repository at `https://github.com/leosakharoff/democracy-stack-paper` and are also submitted as a supplementary archive.

Limitations. The main limitation is that our QA-to-democracy mappings are analytical, not empirically validated through user studies or deployment outcomes. Architectural recovery captures dependency structure and extension mechanisms, not runtime behaviour. Evolutionary metrics use email-based author identity, which conflates authors with multiple emails. Fork analysis covers public forks only. All architectural and ecosystem judgements were made by one coder; the rubric in Table 1 mitigates but does not eliminate inter-rater variance. The two-case

design demonstrates the framework but does not validate it across the wider civic tech field. See Section 6.5 for the full discussion.

4 Framework

This section presents the framework. Section 4.1 introduces its three components: a set of quality attributes mapped to democratic capacities, the three-structures lens for analysing software ecosystems, and a derived set of architectural trade-offs. Section 4.2 describes a separate methodological contribution, fork-divergence shape as a niche-creation diagnostic. Section 4.3 states the procedure for applying the framework to a new platform.

4.1 The Three Components

The three components below offer overlapping but complementary views of a platform: the quality attributes look at the software structure itself, the three-structures lens locates the software within its organisational and business context, and the architectural trade-offs name the choices that follow from the QA mapping.

4.1.1 Quality Attributes Mapped to Democratic Capacities

Each of the four quality attributes introduced in Section 2.1 connects an architecture choice to a democratic capacity the platform can support. Each one also pairs with an architectural cost the platform has to pay if it chooses that direction. Table 2 collects the mappings.

Table 2: Quality attributes mapped to democratic capacities and the architectural costs they impose.

Quality attribute	**Democratic capacity**	**Architectural cost**
Modifiability	Pluralism: diverse communities can adapt the platform	Sustainability burden of N composable parts
Interoperability	Composability: platforms can join wider stacks	Reduced local autonomy under shared standards
Configurability	Procedural pluralism: participation can be organised in different ways	Configuration surface complexity; maintenance overhead
Transparency and auditability	Legitimacy, accountability, and contestability can be supported through inspection and verification	Audit infrastructure: logs, versioning, permissions, and API discipline

The mappings are analytical, not causal proofs. They should not be read as claiming that these quality attributes automatically produce democratic outcomes. Rather, they identify the democratic capacities that each attribute can support when matched by appropriate organisational and institutional conditions. A system whose architecture supports modifiability, for example, creates the conditions for institutional pluralism, but does not guarantee that pluralism will

emerge in practice. We therefore treat the mappings as normative design hypotheses that the application in Section 5 demonstrates against two specific platforms.

Other mappings are possible. Transparency can be read as a precondition for surveillance as well as for accountability; configurability can be read as administrator capture as well as procedural pluralism; interoperability under shared standards can entrench dominant platforms rather than enabling composition. The mappings we use are not the only defensible reading. They are the readings that match our two cases and the democratic capacities the platforms themselves frame their work against. A different corpus or a different normative starting point would land on different mappings, which is part of why we treat this as a small analytical framework rather than a fixed taxonomy.

Configurability and modifiability sit next to each other in the table but address different audiences. Modifiability is about whether the platform can be changed at the code level (a developer's question); configurability is about whether an administrator can compose different participatory processes from the existing components without writing code (an operator's question). Both matter for democratic infrastructure, but they reach different actors and support different democratic capacities.

The framework's four QAs share *Modular Politics*'s structural choice of naming a small set of design-relevant properties [Schneider et al., 2021]. Two of Schneider's principles overlap in name with QAs in this paper (modularity, interoperability), but operate at a different level. Schneider's modularity is software composability at the governance-module level, designed to be usable by platform operators and community members; modifiability in this paper is software modularity at the engine/codebase level, usable by developers. Schneider's interoperability includes governance systems "influencing each other's processes"; ours is closer to standard SA interoperability — the exchange of meaningful information via interfaces [Bass et al., 2021]. The frameworks are also asymmetric on what each leaves out: Schneider's expressiveness and portability are not in our QAs (a limitation we return to in Section 6), and our configurability and transparency/auditability are not in his four (a consequence of his unit of analysis being a proposed governance layer to be embedded in existing platforms, rather than the existing platform architecture itself).

4.1.2 The Three-Structures Lens

The QAs above describe properties of the software structure alone. Christensen et al. [2014]'s three-structures framework gives us a way to think about how the software structure relates to the broader ecosystem.

The three structures — organisational, business, software — are introduced in Section 2.2. Christensen et al. argue that ecosystems thrive when the three fit each other and struggle when they do not.

This lens does two things for us. First, it locates the QAs: the four attributes above are properties of the software structure, but their consequences play out in the organisational and business structures. Modifiability matters for pluralism only if the organisational structure includes

contributors who are able to use the extension mechanisms. Transparency matters for legitimacy only if there are actors (communities, regulators, deployers, researchers) able to inspect and act on what they see. Second, it identifies a failure mode beyond bad architecture: *misalignment* between architecture and the organisational and business reality around it. A modular architecture without a coordinating body to review, integrate, and maintain contributions may produce fragmentation rather than pluralism. A modifiable platform without sustained funding for maintenance may produce decay rather than adaptation.

4.1.3 Architectural Trade-offs

The four QAs cannot all be optimised without cost. Each one pairs with an architectural burden (Table 2), and choosing one direction means accepting that burden elsewhere in the ecosystem.

Three pairings are particularly visible in civic tech and form the framework's architectural trade-offs:

Modifiability against sustainability burden. Modular architectures let many deployers — municipalities, NGOs, civic-tech consultancies — adapt the platform; the cost is maintaining many parts. Monolithic architectures can be easier to maintain centrally; the cost is that adaptation often requires forking or direct source modification unless other extension mechanisms are provided.

Interoperability against reduced autonomy. Interoperable platforms compose into wider stacks; the cost is constraints on local choice under shared standards. Stand-alone platforms preserve local autonomy; the cost is isolation.

Configurability against maintenance overhead. Highly configurable platforms support many different participation processes without code changes; the cost is configuration surface complexity and the maintenance of options across versions. Less configurable platforms are simpler; the cost is that adapting to a new process may require code changes.

Each trade-off is not only an engineering choice but also a governance choice. The architecture makes some forms of adaptation, coordination, and oversight easier than others, whether or not the builders frame the decision politically. The framework makes these choices explicit.

Transparency and auditability is treated separately. Unlike the three above, the cost of transparency (audit infrastructure) does not pull against another democratic capacity directly; it is a cost the platform pays for legitimacy, accountability, and contestability all at once. We discuss it as a primary axis in Section 5 but do not pair it with another QA as a trade-off.

4.2 Fork-Divergence Shape: A Niche-Creation Diagnostic

Fork-mining research has developed several lenses on fork divergence: evolution-pattern typologies of fork–upstream pairs derived from commit-history graphs [Zhou et al., 2020], scalar entropy measures of fork diversity in modified files [Wang et al., 2023], and longitudinal studies of code propagation between mainlines and divergent forks across software ecosystems [Businge et al.,

2022]. Wang et al. [2023] motivate their work with a premise close to ours — that "merely having a large number of forks does not necessarily imply a productive and healthy OSS project" — and Businge et al. [2022] report little code propagation between mainlines and divergent forks across the Android, .NET, and JavaScript ecosystems — and the fork→mainline direction is rarer still — a pattern we replicate in the civic tech setting for Consul. None of this work has been applied to civic tech platforms, and none connects fork behaviour to SECO health dimensions explicitly.

Standard SECO niche-creation metrics, in turn, count contributions but do not by themselves distinguish healthy ecosystem extension from silo formation. We adapt the fork-mining literature into a population-shape diagnostic — *fork-divergence shape* — that reads the distribution of public forks across ahead-of-upstream and behind-upstream states, pairs with Jansen [2014]'s niche-creation dimension, and reads as a three-structures alignment check for civic tech platforms where forks are a relevant extension mechanism.

Operationally, each fork can be located in one of four states, defined by whether it has local commits ahead of upstream and whether it has fallen behind upstream. Table 3 names the four cells and the interpretation we attach to each.

Table 3: The four fork-divergence states. Cell colours match the quadrants of Figure 5, so the table reads as a legend for the figure.

	Behind upstream = 0	**Behind upstream > 0**
Ahead upstream > 0	*Ahead-only* — healthy extension; local work kept current with upstream	*Ahead-and-behind* — diverged silo; local work drifted from upstream
Ahead upstream = 0	*In-sync* — copy that mirrors upstream	*Behind-only* — dormant copy, fallen behind

The shape of the fork population, not the total count, is the diagnostic. Healthy ecosystem extension should populate the ahead-only region: deployers keep up with upstream and add their own work, producing changes that upstream can in principle absorb. Silo formation should populate the behind-only and ahead-and-behind regions: either dormant copies fall behind, or active deployments accumulate local changes while also missing upstream changes. We apply this to Consul in Section 5.2 and find an empty ahead-only region, which we interpret as a fragmentation signal.

4.3 Using the Framework

The framework is applied to a civic tech platform in five steps. First, recover the software structure from primary sources: codebase organisation, gemspec or equivalent manifests, extension mechanisms, APIs, and documentation. Second, identify the organisational structure (who participates and how they coordinate) and the business structure (who pays, who captures value, what incentives flow) from governance documents, funding records, and secondary case studies. Third, rate the four quality attributes against this evidence using the rubric in Table 1 (Section 3); each rating is justified by specific architectural choices or data. Fourth, assess whether the three

structures fit each other: does the software architecture afford the organisational reality, and does the business structure sustain the architectural costs? Fifth, if the platform's deployers use forks rather than plugins or modules, plot the fork population across the ahead-of-upstream and behind-upstream states to read niche-creation health.

The ratings are structured judgements rather than measurements.

The output is a structured comparison the framework's tables can hold (Table 2 for the QA profile, Table 4 for the three structures). The next section walks through this procedure for Decidim and Consul.

5 Application: Decidim and Consul

This section applies the framework to Decidim and Consul. For each platform we describe the three structures (organisational, business, software), profile the four quality attributes, and connect the observations to the trade-offs and the architecture-as-governance thesis. The analysis treats repository data as indicators of activity, concentration, and divergence, not as direct measures of democratic value. A cross-case synthesis follows in Section 5.3.

5.1 Decidim

Decidim is a modular monolith built with Ruby on Rails. The codebase at `github.com/decidim/decidim` contains 27 first-party engines packaged as standalone Ruby gems at SHA `be39c244` (2026-05-12).

Software structure. The 27 engines fall into four groups: `decidim-core` as the foundational dependency, three infrastructure engines (`admin`, `api`, `system`), three shared-service engines (`comments`, `forms`, `verifications`), four participatory spaces (`processes`, `assemblies`, `conferences`, `initiatives`), and 13 components that attach to those spaces. Every engine except `decidim-core` declares `add_dependency "decidim-core"` in its gemspec, matching a hub-and-spoke pattern.

However, 13 of the 27 engines have runtime dependencies on other Decidim engines besides core: `decidim-comments` has 6 user-facing dependents and `decidim-forms` has 5, making them Layer-2 hubs of the platform (Figure 1).

The key architectural pattern is the space/component matrix: almost any component can attach to almost any space, creating an $N \times M$ combinatorial structure implemented through a polymorphic database association in core. Barandiaran et al. [2024, p. 82] call this the "furniture" model: administrators mix and match components with spaces without writing code. Components are registered through a manifest API (`Decidim.register_component`); third-party modules use the same interface as first-party ones.

To examine whether modular design is reflected in modular evolution, we applied Gall et al.'s logical coupling analysis [Gall et al., 1998] to the commit history. Of 5,517 commits (excluding mega-commits per Section 3), 64% stay within a single engine. This suggests that the engine

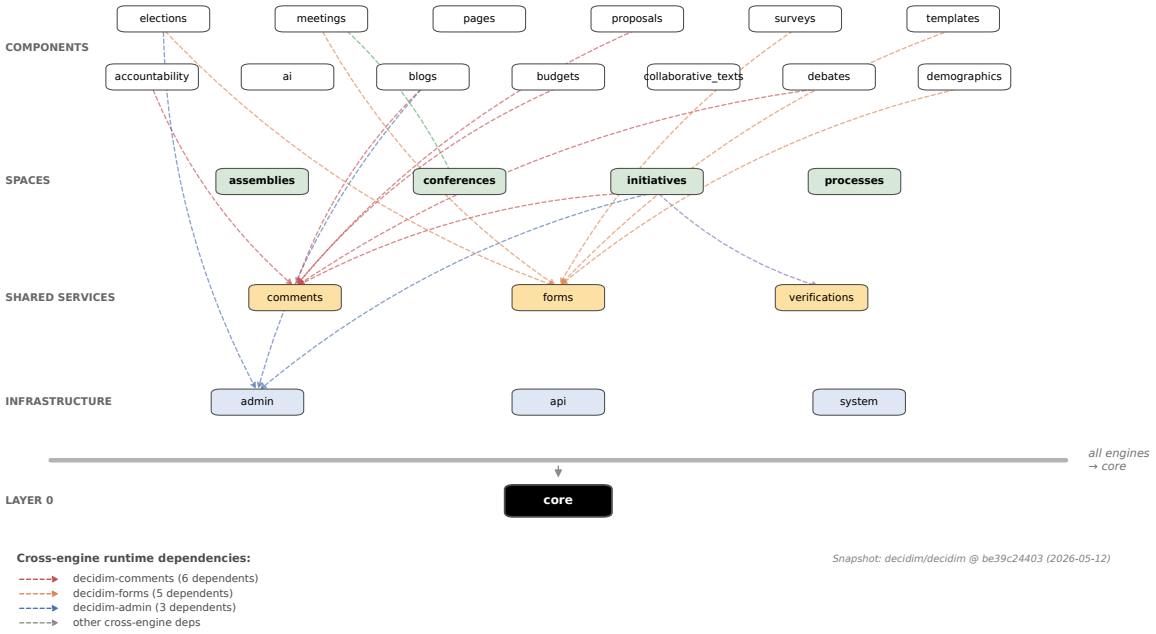


Figure 1: Decidim engine dependency graph at SHA `be39c244` (2026-05-12), recovered from the 27 gemspec files. The grey bar above `decidim-core` stands in for the 26 universal core dependencies. The dashed arrows are cross-engine runtime dependencies that bypass core; the most-depended-on are `comments` (6 dependents) and `forms` (5). Three developer-tooling engines (`design`, `dev`, `generators`) are omitted.

architecture provides meaningful change isolation at the repository level, although it does not prove runtime independence. The remaining cross-engine coupling is concentrated in the four participatory spaces (Jaccard up to 0.43), which share infrastructure code despite having no gemspec dependencies between them (Figure 2).

Organisational structure. The Decidim Free Software Association was constituted on 16 February 2019 in a community assembly that approved its statutes. The Association holds the source code and trademark, transferred from Barcelona City Council in 2017. Its governance includes a General Assembly as sovereign body and four standing committees: Coordination, Research, Decidim Politics, and Public Institutions. The Public Institutions Committee brings Barcelona City Council, the Generalitat de Catalunya, the Barcelona Provincial Council, Localret, and other public bodies into the coordination structure [Barandiaran et al., 2024].

MetaDecidim is the community layer: a public Decidim instance at meta.decidim.org through which both governance and product work happen. The Decidim Social Contract, binding on members, commits signatories to AGPLv3 licensing, transparency, and "equal opportunities for participatory objects" — the no-ranking-bias rule that surfaces in the software as a randomised default sort. The Association runs a small technical office (three employees per Barandiaran et al.) that handles release management and code review. Module ownership is distributed across Codegram engineers, Association staff, and the wider contributor base; 183 unique authors are visible in the commit log.

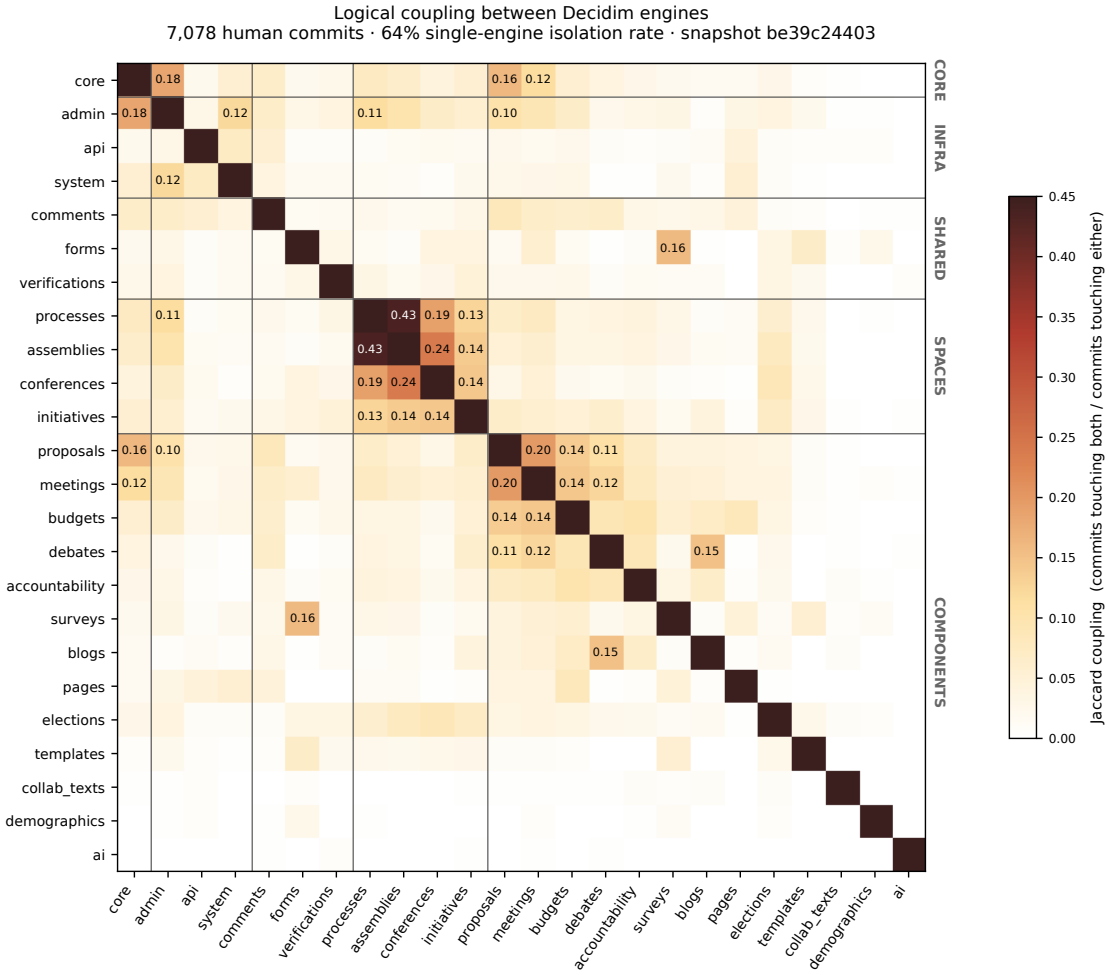


Figure 2: Logical coupling between Decidim engines, computed as Jaccard similarity over the project's commit history at SHA `be39c244`. The four participatory spaces (centre block) co-change despite having no gemspec dependencies between them. The shared services (`comments`, `forms`, `verifications`) are largely isolated — their gemspec-dependents call them as APIs without co-changing.

Business structure. The Association is funded through annual transfers from Barcelona City Council and the Generalitat de Catalunya under a formal collaboration agreement, with additional support from the Consorci Localret and the Barcelona Provincial Council [Barandiaran et al., 2024]. Around this institutional core sits a network of anchor consultancies that fund themselves through municipal contracts: Codegram (Barcelona), Octree (Geneva), Open Source Politics (Paris), and smaller partners. The economic loop is: municipalities procure deployments from these consultancies; the consultancies are paid for installation, customisation, hosting, and bespoke modules; some of the work can flow back upstream as gem-based community modules or core pull requests. Decidim documents more than 80 community-developed gems, a visible artefact of this return flow.

Fork divergence. Of 452 enumerated forks of `decidim/decidim`, 444 had upstream comparison data. The distribution is concentrated in the behind-only quadrant: 399 forks (90%) have no local commits but have fallen behind upstream as Decidim ships on `develop`. Forty-five forks sit in the ahead-and-behind quadrant; none in ahead-only or in-sync. The most-divergent fork is 221 commits ahead — a Spanish-language rebrand of the documentation website. Consul's deepest silo, by contrast, is 12,725 commits ahead. Two orders of magnitude separate the two cases.

The originator pattern inverts. Madrid (Consul originator) abandoned a 12,725-commit fork in 2023. Barcelona (Decidim originator, `AjuntamentdeBarcelona/decidim`) sits at zero commits ahead and forty-six behind, with a push timestamp from the day before the snapshot — the originator is tracking upstream. Codegram, the consultancy that built Decidim, also sits at zero commits ahead. AjuntamentManacor and other municipal forks show the same pattern. Figure 3 plots all 444 comparable forks on identical axes to Figure 5, so the contrast reads directly.

This is not a claim that Decidim has a healthier deployment ecosystem in raw numbers. The documented installation path is `gem install decidim; decidim my_app`, which produces a new Rails application living in its own repository rather than as a fork of upstream. Most Decidim deployments are therefore invisible to fork-tree analysis. The 444 fork population captures contributors with PRs in flight (typically single-digit ahead counts), learners reading the codebase (zero ahead, accumulating behind), and a small minority of deep-customisation deployers. The displacement of the deployment fork by the plugin path — not the small ahead-and-behind cluster — is the empirical signal.

Quality attribute profile. *Modifiability* is high. The engine architecture lets deployers add, remove, or replace components without touching core code; customisation happens through modules, not deployment forks. The 80+ community modules are evidence that this modifiability is used in practice.

Interoperability is moderate. A GraphQL API exposes most participatory content, and the project's integration choices (OpenStreetMap, Matomo, Jitsi) are deliberately open-source. There are no cross-platform standards for civic tech that would let Decidim compose with other platforms at the data layer, so interoperability remains less developed than modifiability or configurability in this case.

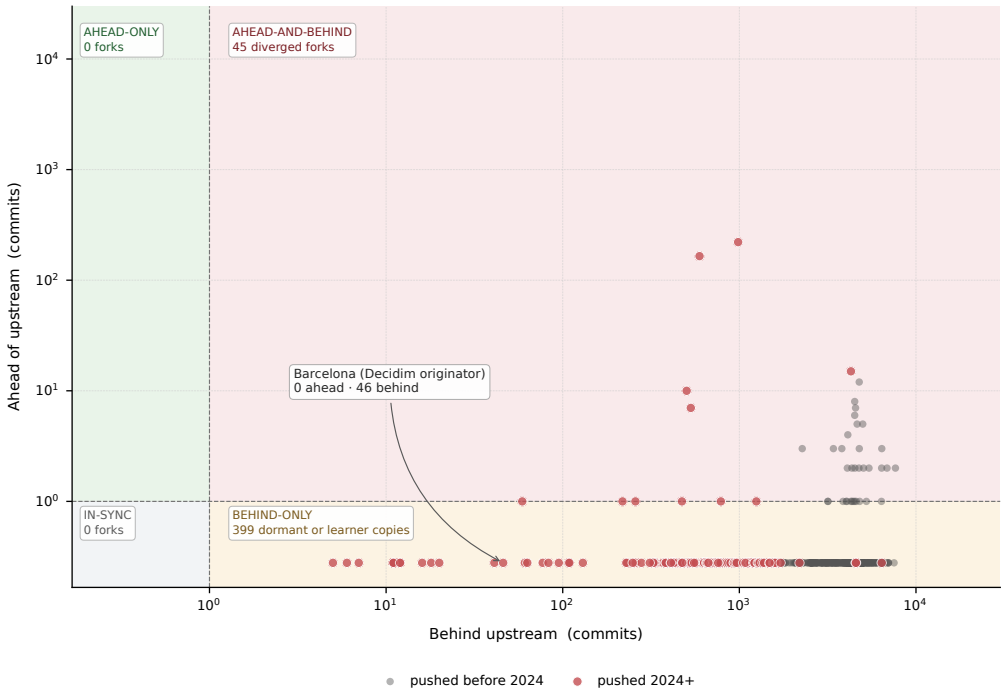


Figure 3: Decidim forks plotted by commits ahead of upstream `develop` against commits behind (snapshot 2026-05-14), on the same axes as Figure 5. The ahead-only region is empty, as for Consul. The mass sits in the behind-only quadrant (399 dormant or learner copies, 90%). The ahead-and-behind cluster is small in both count (45 forks) and magnitude (top fork at 221 commits ahead, two orders of magnitude below Consul's Madrid). Barcelona, the Decidim originator, sits at zero ahead and forty-six behind — the inverse of Madrid's silo. Of 452 enumerated forks, 444 had upstream comparison data.

Configurability is high. The 4 spaces and 13 components produce many possible default configurations, each customisable through component-level settings without code changes. Palacin et al. [2024] document 25 distinct production configurations across studied deployments.

Transparency and auditability is high at the architectural level. Decidim ships an immutable `ActionLog` model with a four-level visibility enum (private, admin-only, public-only, all) and an organisation scope. The `Traceable` concern adds PaperTrail-based version history to every changeable model, so per-proposal version diffs are available. Permission-gated controllers enforce admin-log access. The GraphQL API is designed to expose public participatory content while not exposing individual votes or admin actions. Sorting is deterministic and seeded for reproducibility within a session. These mechanisms make auditability a first-class architectural concern, although they do not by themselves prove that non-technical citizens can meaningfully use the audit surfaces in practice.

Figure 4 plots per-engine commits in the 12 months before the snapshot. All 24 user-facing engines see activity, with three new engines (`ai`, `demographics`, `collaborative_texts`) accumulating their first hundreds. The modular architecture is therefore matched by visible maintenance activity across the engine set; no user-facing engine is dormant in this snapshot.

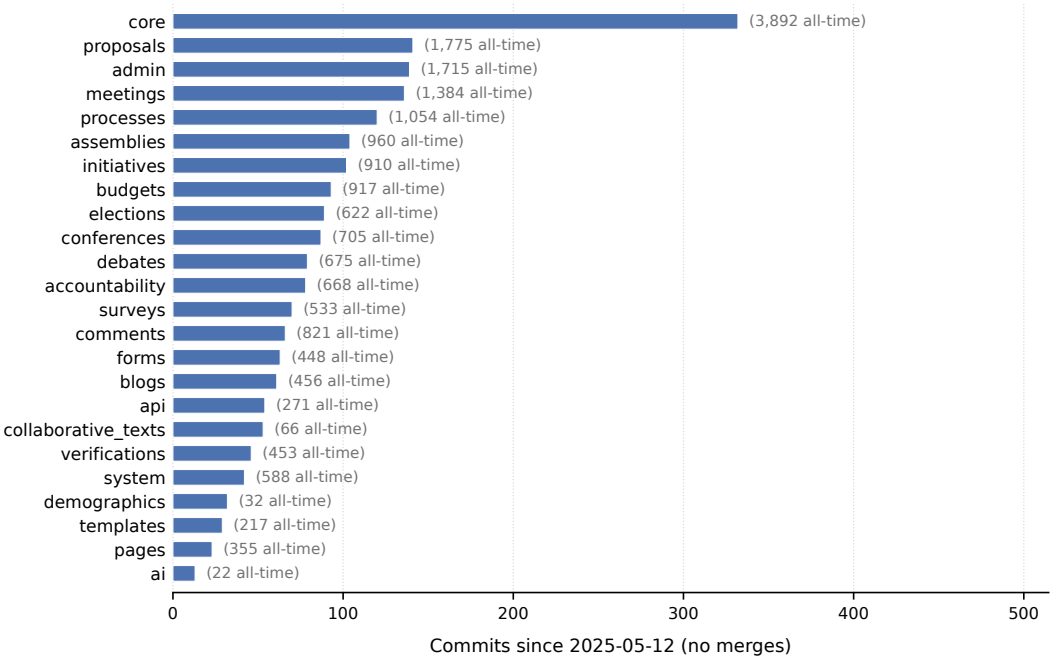


Figure 4: Decidim commits per user-facing engine in the 12 months before the snapshot (2025-05-12 to 2026-05-12). Bar values are commits in the period; the annotations show all-time commits for context. Three developer-tooling engines (`design`, `dev`, `generators`) are excluded, matching Figure 1.

Three-structures alignment. The three structures appear aligned. The modular software lets external contributors ship gems without negotiating commit access to the core repository; the Association coordinates contributions and runs keystone functions such as releases, code review, and integration; the consultancy network has a business model around incremental deployment and module work; and the Public Institutions Committee channels public-body requirements

back upstream. When Barcelona pays for accessibility work, that work can land in `decidim-core` and become available to other municipalities, which is the commons logic Barandiaran et al. articulate. The 80+ community modules and the steady commit rate after 2019 are visible indicators of this alignment.

5.2 Consul

Consul and Decidim share an origin in Spain's 15M/Indignados movement. Barcelona initially contributed to Consul but proposed modularising the codebase. The proposal was rejected; Barcelona rewrote from scratch, creating Decidim in 2016–2017 [Codegram, 2024]. The original repository has since been renamed from `consul/consul` to `consuldemocracy/consuldemocracy`. Recovery is pinned to SHA `3c3b5c4a` (2026-05-11).

Software structure. Consul is a single Rails application: no gem separation, no engine isolation. Features (proposals, debates, budgets, legislation, voting, polls) share the same codebase, the same `ApplicationController`, and one `config/routes.rb`. Models live in `app/models/` with functional subdirectories but no module-level namespace boundaries enforced. All tables sit in one schema. There is no `ARCHITECTURE.md`, and the technical documentation covers installation but not architecture.

Customisation uses a "custom shadow folder" pattern: to change how a budget investment works, a deployer creates a file at the same path under `app/models/custom/` that overrides the original. This reduces some file-level conflicts but does not provide module isolation. The documentation's "Getting Started" step 1 is "Create your fork", making deployment forks part of the expected customisation path.

Organisational structure. The Consul Democracy Foundation was founded in 2019 as a Dutch non-profit [Linåker and Gates, 2025]. It was set up specifically because Madrid's institutional backing for Consul collapsed after the May 2019 municipal election. The Foundation is very small: a Director, a Network Coordinator, a Tech Coordinator, and a three-person board. Crucially, the Foundation does not employ developers. Linåker and Gates [2025] report that maintenance relies on "a technical core group dominated by Spanish service suppliers dating to Consul's early days." The community is held together through annual ConsulCon events, a Slack workspace of 700–800 users, and a two-tier Certified Tech Partner programme. There is no Decidim-style Social Contract, no Public Institutions Committee, and no MetaDecidim-equivalent space where deployers participate in governance.

Business structure. The Foundation is funded by a patchwork: a single municipal sponsor (the City of Munich at EUR 20,000 per year), EU project grants, private donations, and revenue contributions from Certified Tech Partners [Linåker and Gates, 2025]. The Foundation does not commission core development directly; it cannot afford to. Municipalities procure Consul deployments from certified suppliers, but the architecture means much customisation happens inside the deployer's own fork rather than as a returnable contribution. Linåker and Gates [2025] report that contributions "come largely from certified service suppliers, with fewer contributions

from local governments and NGOs" and that "many development features rarely merge back to the core codebase due to architectural barriers and fragmentation across installations."

Fork divergence. We enumerated all forks of `consuldemocracy/consuldemocracy` via the GitHub API and computed each fork's divergence against upstream master. Of 1,030 enumerable forks, 1,026 had upstream comparison data. 746 (73%) are unmodified "stars in fork form"; 280 (27%) have at least one commit ahead — approximately the "around 250 deployments" figure previous reports cite [NTARI, 2025]. The distribution within those 280 is heavily skewed: most have fewer than ten commits ahead, 39 have 100 or more, and only 5 have 1,000 or more. Activity has been declining: pushes peaked in 2019 and only 14% of forks have been touched since the start of 2024 (Figure 6).

The most-diverged fork is `AyuntamientoMadrid/consul` — Madrid itself, the city that built Consul. It sits 12,725 commits ahead of upstream and 11,540 behind, last pushed 2023-05-23. The originating city therefore maintained a separate copy that drifted far from upstream and was eventually abandoned. This does not prove that the architecture alone caused the divergence, but it shows that even the originator ended up using the fork pattern that the platform made easiest. Plotted in Figure 5, three quadrants are populated and one is conspicuously empty: 280 ahead-and-behind silos (top-right), 745 behind-only dormant copies (bottom-right), 1 in-sync fork (bottom-left), and zero ahead-only forks (top-left). The empty top-left is the diagnostic signal — no fork keeps up with upstream while also adding local work.

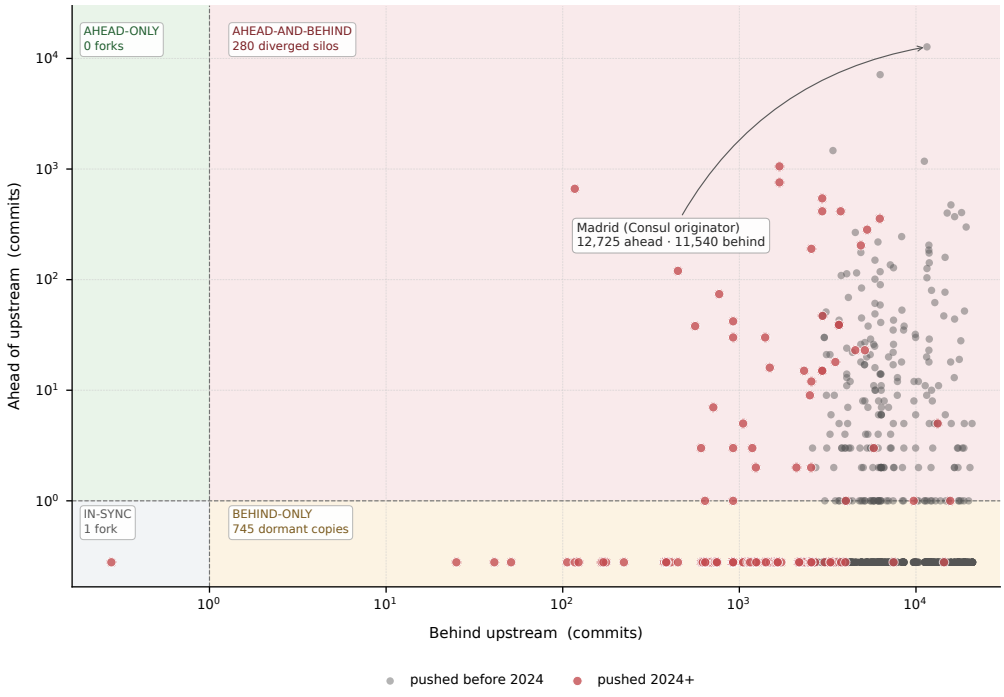


Figure 5: Consul forks plotted by commits ahead of upstream master against commits behind, on log scales (snapshot 2026-05-12). Each fork occupies one of four quadrants. The ahead-only region — where healthy contributor forks would live — is empty: no fork has added local commits while staying in sync with upstream. The mass sits in the behind-only region (745 dormant copies) and the ahead-and-behind region (280 diverged silos, including Madrid). Of 1,030 enumerated forks, 1,026 had upstream comparison data; the remaining four are excluded.

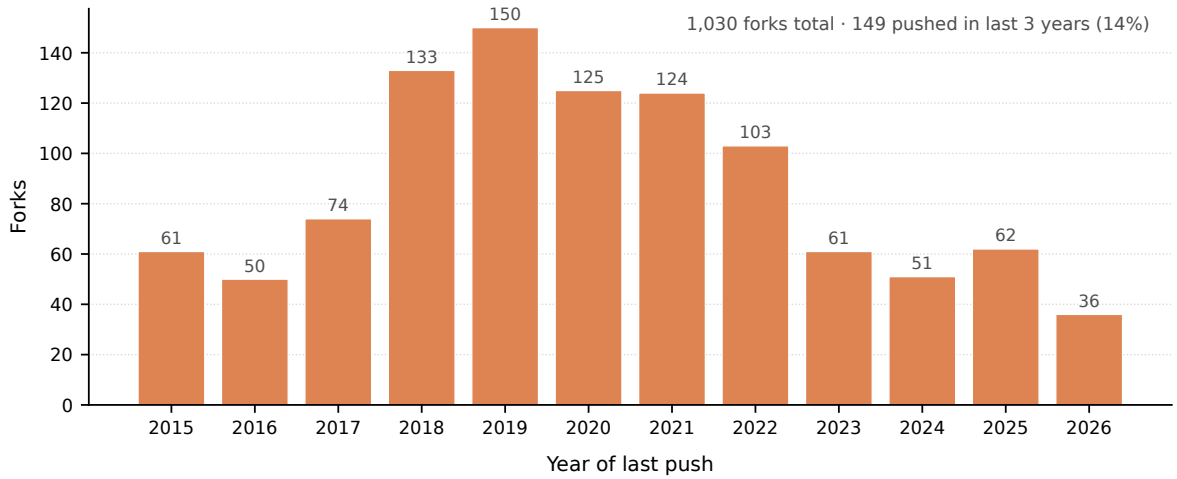


Figure 6: Last-push year distribution for the 1,030 enumerable Consul forks (snapshot 2026-05-12). Each bar counts forks whose most recent push falls in that year. Activity peaks in 2019 and declines; only 14% of forks have been touched since the start of 2024.

Quality attribute profile. *Modifiability* is low. Customisation is expected to happen through forks and shadow-folder overrides rather than through isolated modules. Shadow folders reduce file-level conflicts but do not remove logical coupling to the upstream codebase. The 1,030 forks are the visible signature of this extension model, although not every fork represents a production deployment.

Interoperability is moderate. The public GraphQL API at `/graphql` exposes participatory content including individual vote records, but the data model is monolithic and there are no cross-platform standards.

Configurability is moderate. Several components (proposals, debates, budgets, voting) can be enabled or disabled per deployment, with phase- and quorum-level settings. The configuration is less rich than Decidim's space/component matrix.

Transparency and auditability is moderate. Consul uses the off-the-shelf `audited` gem to track per-record changes, but the UI exposes audit history only for Budget Investments. A separate `Activity` model logs five specific moderation verbs (hide, block, restore, valuate, email). Both surfaces are admin-only. There is no Decidim-style organisation-scoped `ActionLog` with public visibility options.

Three-structures misalignment. The three structures appear misaligned. The problem is not monolithic architecture as such: a monolith can work well when it is matched by strong central maintenance and clear contribution pathways. The problem is the fit between Consul's monolithic extension model and its organisational and business structures. The software structure makes local adaptation easiest through forks; reintegrating that work would require a strong coordinating body with sustained development capacity; the small Foundation cannot provide that capacity; and the business structure does not reliably pay anyone to provide it. The architecture's costs therefore land on individual deployers, who absorb them by maintaining

isolated forks. The misalignment shows up in the fork data: 1,030 forks, 750 unmodified copies, and 280 modified forks with limited evidence of upstream return.

Two pieces of evidence make this concrete. First, Madrid itself ended up in the same fork pattern as other deployers, even with the resources and motivation of the originating city. Second, the post-2019 collapse: when Ahora Madrid lost the mayoralty in May 2019, the upstream commit rate dropped roughly 80% within a year. The Foundation was constituted that same year specifically to fill the institutional vacuum, but without core developers or significant funding it could coordinate the ecosystem more easily than it could maintain the core. The hedge here matters: the claim is not that the monolith caused the misalignment, but that this monolithic extension model required a coordinating capacity that the organisational reality could not provide.

5.3 Cross-case Synthesis

Decidim's contributor base is less concentrated than Consul's, and Consul's upstream activity collapsed in 2019. The commit history makes both patterns visible. Figure 7 shows the concentration: Decidim has 7,078 human commits across 183 unique authors with a Gini coefficient of 0.85 (top-three share 36%, top-ten 68%), against Consul's 186 unique authors with a Gini coefficient of 0.93 (top-three 52%, top-ten 87%). These figures indicate contributor concentration in the repository, not the full distribution of influence in the wider ecosystem. Figure 8 shows the Consul cliff in 2019–2020 following Ahora Madrid's electoral loss in May 2019 [Linåker and Gates, 2025, Barandiaran et al., 2024], alongside Decidim's steadier plateau after 2019.

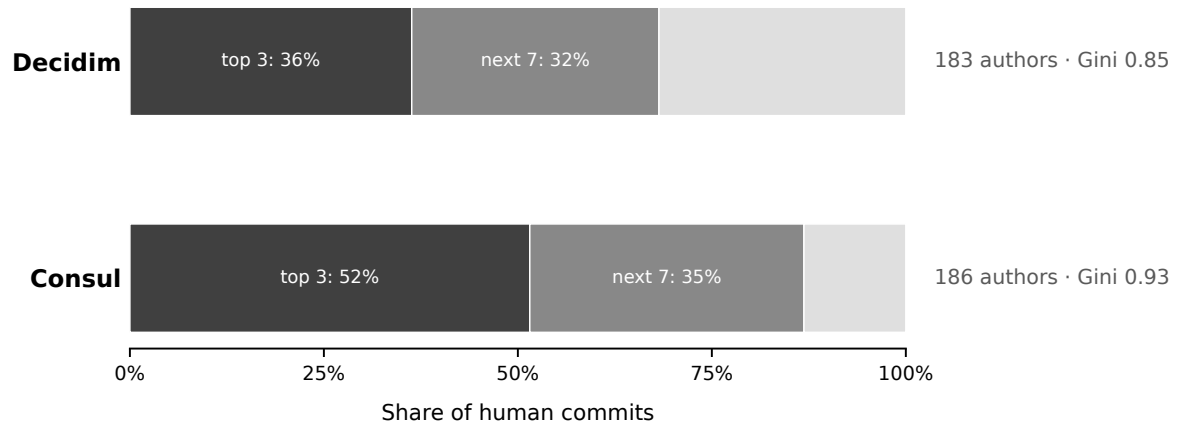


Figure 7: Share of human commits captured by the top three and top ten authors per repository. Decidim has a longer tail of unique contributors; Consul is more concentrated. Bot commits (Dependabot, Crowdin, etc.) are excluded.

Three-structures comparison. Table 4 maps the two cases onto the three-structures framework.

Decidim's structures appear aligned. The modular software affords incremental contribution; the Association coordinates it; the consultancies help sustain it. Consul's structures appear

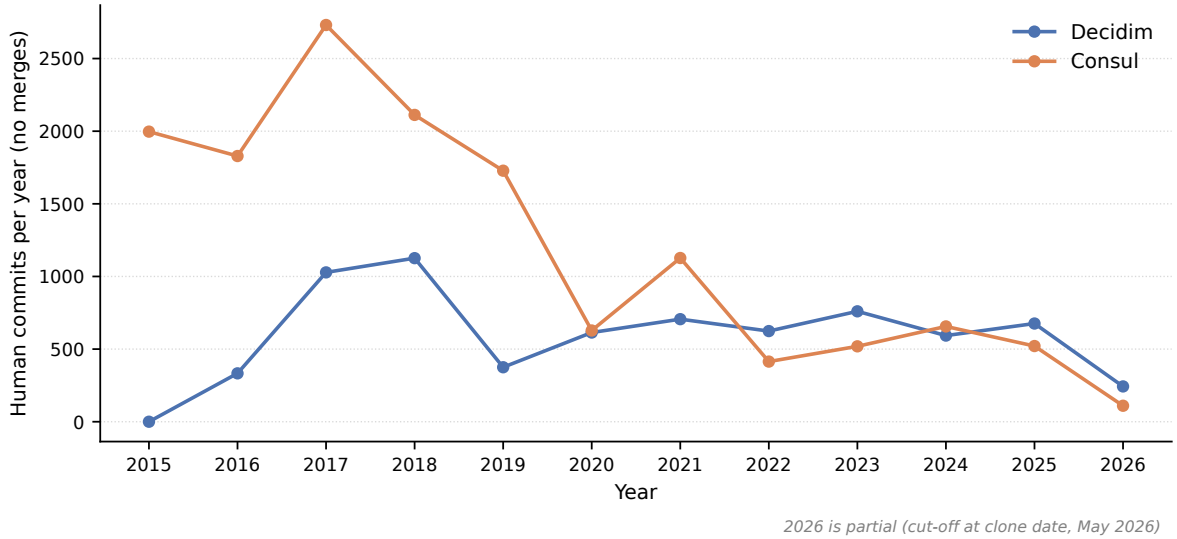


Figure 8: Human commits per year for each repository (no merges, bots excluded). The Consul cliff in 2019–2020 follows Ahora Madrid's electoral loss in May 2019; Decidim shows a steadier plateau after 2019. The 2026 numbers are partial (cut-off at clone date).

Table 4: Three-structures analysis [Christensen et al., 2014] applied to Decidim and Consul.

Structure	**Decidim**	**Consul**
Software	Modular: 27 engines, registry pattern, shared DB, gem-based extension	Monolithic: single Rails app, shadow folder pattern, fork-based extension
Organisational	Decidim Association as keystone; MetaDecidim community; Social Contract; Public Institutions Committee	Consul Democracy Foundation (6 people, no developers); informal supplier network; weak coordination
Business	Catalan institutional funding + anchor consultancies (Codegram, Octree, Open Source Politics) + Public Institutions Committee feedback loop	EUR 20,000/yr from Munich + EU grants + supplier certification revenue; no anchor consultancy

misaligned. The monolithic extension model requires a strong coordinator that the Foundation cannot afford to be, and deployers absorb the misalignment by maintaining forks.

The fork-divergence figures (Figures 5 and 3, on identical axes) make this contrast visible. Madrid, the Consul originator, sits in the far corner of the ahead-and-behind quadrant with a 12,725-commit fork abandoned in 2023. Barcelona, the Decidim originator, sits at zero local commits with a fork pushed the day before the snapshot. The largest divergent Decidim fork is 221 commits ahead — two orders of magnitude below Consul's Madrid. The same diagnostic reads each platform's structural alignment empirically: the architecture that pushed Consul's deployers into silos is absent from Decidim, and the originator pattern on each side fits the structural prediction.

Trade-off positioning. The two platforms sit at opposite ends of the modifiability trade-off identified in Section 4.1.3. Decidim chose high modifiability and pays the sustainability burden of a 27-engine codebase maintained by a small technical office. Consul chose lower modifiability and pays the cost in fork fragmentation: 280 modified forks with little visible upstream return. On interoperability, both are open-source and have GraphQL APIs but neither pursues cross-platform civic tech standards; the trade-off does not differentiate them strongly and remains under-tested in this two-case application. On configurability, Decidim's space/component matrix is high and Consul's per-feature settings are moderate; the cost on the Decidim side is the configuration complexity Palacin et al. [2024] document.

Transparency. Transparency and auditability differ sharply across the two at the architectural level. Decidim treats auditability as a first-class architectural concern through the `Action-Log/Traceable` infrastructure; Consul treats it more narrowly as a moderation oversight feature implemented via off-the-shelf gems. This differentiation supports including transparency and auditability as a primary axis of the framework, while leaving open the separate question of how usable these mechanisms are for non-technical citizens.

6 Discussion

6.1 Architecture as Governance in Practice

The Decidim/Consul comparison shows how architectural decisions participate in governance in civic technology. Decidim's modular architecture did not only make the code flexible; it also made it possible for communities and service providers to shape the platform without first negotiating commit access to the core repository. An organisation can build a module, publish it as a gem, and have it work alongside the official engines. This is close to what Schneider et al. [2021] call "modular politics": composable governance primitives implemented as software. The 80+ community modules are one visible artefact of this politics in action.

Consul's monolithic extension model made a different path easier. The fork-and-shadow pattern made isolated local adaptation more straightforward than upstream contribution: deployers could either accept the platform as-is or maintain a separate fork. Schneider [2024] argues that

platform defaults encode governance; in Consul's case, the default extension path pulled local adaptation toward forked copies rather than shared modules. The pattern reached its most visible case in Consul's own originator: Madrid maintained a fork 12,725 commits ahead of upstream, last touched in May 2023. This does not prove that architecture alone caused the fragmentation, but it shows that even the originator ended up in the same divergence pattern visible across the wider deployer base.

The two platforms can also be read through Iansiti and Levien [2004]'s ecosystem actor roles. Decidim's Association operates as a keystone, maintaining shared infrastructure while enabling third-party modules on top; the 80+ community modules are the niche-creation activity this keystone structure makes possible. Consul's Madrid team operated more like a dominator while the city's institutional backing held: it controlled the codebase, rejected the modularisation path proposed by Barcelona, and offered no equivalent module mechanism for external contribution. When the institutional backing collapsed in 2019, no successor keystone with comparable development capacity emerged. The vacuum was partly absorbed by deployers maintaining their own versions in forks.

6.2 What SECO Research Gains

The SECO health dimensions from Jansen [2014] — productivity, robustness, niche creation — offer useful organising concepts for civic tech, but they do not by themselves distinguish ecosystem extension from silo formation. Consul's 2017–2019 peak generated thousands of commits and dozens of unique contributors — numbers a productivity-only assessment might read as a thriving project. Yet the extension model pushed adopters toward separate forks, and when Ahora Madrid lost the city in 2019 the upstream commit rate dropped roughly 80%. By productivity measures, the ecosystem looked active; by fork-divergence measures, it was already vulnerable to fragmentation.

Fork-divergence shape as a niche-creation diagnostic (Section 4.2) is one concrete methodological contribution this paper makes to SECO research, adapting the fork-mining literature [Zhou et al., 2020, Wang et al., 2023, Businge et al., 2022] into a population-shape reading paired with Jansen [2014]'s niche-creation dimension for civic tech platforms. Figures 5 and 3 plot the two cases on identical axes. Consul's quadrant pattern — 280 ahead-and-behind silos and 745 behind-only dormant copies, with the ahead-only quadrant empty — is a fragmentation signal: the deployer fork population extends away from upstream rather than around it. Decidim's shape is structurally different: the ahead-and-behind cluster is small in both count (45 forks) and magnitude (top fork at 221 commits ahead, two orders of magnitude below Consul's Madrid), and the originator sits at zero local divergence. The shapes are qualitative and visual, complementary to scalar measures used in the fork-mining literature.

Applying the same diagnostic to both platforms surfaces a methodological point. The shape reads different ecosystem niches in each case, and the contrast is itself the finding. For Consul, where deployers customise by forking, the fork population is approximately the deployment ecosystem and the shape reads deployment fragmentation directly. For Decidim, where deployers install the gem in their own Rails application (`decidim my_app`), most deployments live outside

the fork tree and the shape reads contributor flow plus learners instead. The near-absence of large diverged forks in the Decidim figure is therefore not merely a healthier-looking deployment ecosystem in raw numbers; it is evidence that the plugin path has displaced the deployment fork. Whether fork-divergence shape can be applied uniformly across civic tech platforms thus depends on the documented extension model, and that bound is itself a useful methodological finding.

The three-structures framework of Christensen et al. [2014], applied here to civic tech, also extends the SECO toolkit. The misalignment reading of Consul's fork fragmentation makes the structural argument concrete: not that the architecture was simply bad, but that the extension model required a coordinating capacity the organisational and business structures could not provide. Architecture and organisation diverged regardless of which way the causal arrow runs.

6.3 Implications for the Democracy Stack

International IDEA [2026] calls for a democracy stack but leaves the architectural decisions implicit. CEPS [2025] advocates for interoperable European digital public infrastructure but does not engage the modifiability-against-sustainability trade-off that shapes whether interoperable infrastructure can actually be adapted by the communities it serves. Two lessons from our cases speak to these gaps.

Modularity as a condition for pluralistic adaptation. Trade-off 1 (modifiability against sustainability burden) makes the strongest case for modular architecture in a democracy stack. The Decidim/Consul comparison shows what can happen on either side. The trade-off is real: modular maintenance is harder than maintaining a smaller central codebase. But the alternative, when local adaptation is needed and no other extension mechanism exists, is fork fragmentation that upstream cannot easily absorb. For a democracy stack that must accommodate diverse institutional contexts, modularity appears to be a central architectural condition for pluralistic adaptation. The sustainability burden should therefore be treated not as an implementation inconvenience, but as a governance cost that must be deliberately funded and coordinated.

Transparency by design, not by audit. Decidim's transparency and auditability come from architectural decisions made early: a first-class `ActionLog`, the `Traceable` concern wired into changeable models, and a GraphQL API designed to expose public participatory content while protecting sensitive data. Consul's transparency mechanisms are narrower and implemented mainly through off-the-shelf audit and activity-tracking features. For a democracy stack that has to remain auditable across deployments and over time, transparency is difficult to add as a superficial feature after launch; it has to be treated as an architectural property. The cost — audit infrastructure across models, permissions, visibility levels, and APIs — is real, but without it the legitimacy of participatory outcomes is harder to verify.

6.4 Quality Attributes Not in the Framework

Three quality attributes that matter for democratic infrastructure are not primary axes in our framework: security, accessibility, and sovereignty. We left them out because the Decidim/Consul

comparison does not strongly differentiate the platforms along these axes. Both follow standard Rails security practices; both have accessibility statements and basic WCAG compliance work; both are AGPL-licensed and self-hostable. Engaging these QAs properly would need either different methods (threat modelling for security; user studies for accessibility) or different cases (a commercial SaaS for sovereignty; a federated or anonymous platform for security). They belong in a fuller framework. We name them here to be explicit about what our two cases can and cannot teach.

Two of Schneider et al. [2021]'s principles for *Modular Politics* — expressiveness and portability — are also absent from our framework, but for a different reason: they are normative design goals for a future governance layer (a proposed standard plus supporting libraries to be embedded in existing platforms), not measurable properties of the existing platform architecture itself. Whether a participatory budgeting tool can move from Decidim to Consul is a legitimate civic-tech question, but it requires a different unit of analysis than the architectural recovery this paper does. They belong in a research programme that takes Schneider et al.'s proposal seriously and asks how existing civic tech platforms could support such a layer.

6.5 Limitations

The two-platform comparison is the framework's first worked application, not a validation across the field. Several limitations bound what we can claim.

Causality. The architecture-as-governance reading is correlational. Barcelona and Madrid had different governance cultures and political trajectories; Madrid's Ahora Madrid government lost power in 2019, hurting Consul's institutional backing. Separating what the architecture caused from what politics caused would need interview research we did not do. The hedge is real: the architecture and the organisational reality diverged in opposite directions, and the misalignment matters regardless of which way the causal arrow runs.

Single coder. The architectural and ecosystem judgements were made by one author; a second coder could disagree on edge cases (which engine counts as "shared service", which fork counts as "substantively modified"). Where possible we triangulated against published characterisations — Linåker and Gates [2025] on Consul's organisational structure, Barandiaran et al. [2024] on Decidim's governance, the OSOR and NTARI reports on Consul deployments — and noted disagreements explicitly.

Anglophone bias. Almost all our sources are English; Decidim has substantial Spanish- and Catalan-language documentation, and the Consul fork landscape is heavily Spanish, Catalan, and Latin American. We likely missed material that would refine our reading.

Selection on the dependent variable. We chose two platforms that exist; the civic tech graveyard — platforms that never reached production or failed invisibly — is invisible to this study. The trade-offs we identify are conditional on survival.

Architectural recovery limits. Architectural recovery captures dependency structure and extension mechanisms, not runtime behaviour. Evolutionary metrics use email-based author identity, which

conflates authors with multiple emails. Fork analysis covers public forks only and may miss private production deployments. Each of these is a real limit on what the methods can see.

7 Conclusion

Decidim and Consul started from similar political and historical origins but developed into different software ecosystems. Decidim's modular software appears to fit its coordinating Association and consultancy-anchored funding base. Consul's monolithic extension model appears less well matched to its small Foundation and shoestring funding, and this misalignment is visible in the platform's fork fragmentation. This paper makes sense of that contrast by bringing together three established literatures: software architecture quality attributes, Christensen et al. [2014]'s three-structures lens for software ecosystems, and the architecture-as-governance thesis from STS and platform studies. The contribution is not to invent those components, but to combine them into an analytical framework for civic technology platforms and demonstrate how the framework reads two worked cases. The main methodological contribution that emerged from the application is *fork-divergence shape*: a diagnostic for whether a civic tech ecosystem extends around a shared upstream or fragments into diverged forks.

We applied the framework to Decidim and Consul: two civic technology platforms with shared origins in Spain's 15M movement and different architectural paths. Decidim's modular software, coordinating Association, and consultancy-anchored funding form a structurally aligned ecosystem in this reading. Consul's monolithic extension model, weak coordinating body, and limited funding form a structurally misaligned ecosystem whose fragmentation is visible in fork divergence. The framework reads both cases consistently by asking not only what the software architecture looks like, but whether the organisational and business structures can sustain the architectural choices the platform has made.

This is a foundational step rather than a full validation. A fuller framework would need to examine quality attributes the Decidim/Consul comparison does not strongly differentiate — security, accessibility, and sovereignty — and would benefit from more cases that evaluate the framework against a wider design space.

Future research. Two directions invite follow-on work and could each ground a master's thesis. First, extending the framework to a wider set of platforms: CitizenLab to examine sovereignty under commercial SaaS models, Loomio to examine worker-cooperative governance, and a federated platform to examine security and identity. Second, stakeholder interviews with Decidim deployers, Consul fork maintainers, and Decidim Association members would complement the repository-mining done here with practitioner evidence and address the causal limitations in Section 6.5.

Use of Generative AI

Tool used: Claude (Anthropic; Claude Sonnet 4.6 and Claude Opus 4.7), accessed via the Claude Code command-line interface.

Use in the project: Generative AI was used as an assistant during the research and writing process. It supported literature discovery, outlining and structural decisions, language refinement and copy editing, exploratory web research, and the drafting and debugging of Python and matplotlib scripts used to collect, process, and visualise repository data. The scripts are included in the project repository together with the paper source.

Author contribution and validation: The author developed the research question, case selection, architectural framing, quality-attribute analysis, interpretation of findings, prose, and final argumentative structure. AI was used as a research and writing assistant — supporting refinement of drafts, language editing, code review, and exploration of alternative framings — with all AI-generated content reviewed, revised, tested, and validated by the author against primary sources, public documentation, and repository data. The analytical claims, figures, and conclusions presented in the paper are the author's own responsibility.

Data, copyright, and confidentiality: No confidential material or personal data was provided to the AI tool. The work was based on the author's notes, public documentation, public repositories, and publicly available sources.

All submitted text, figures, code, and analysis were reviewed, edited, and approved by the author.

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